

AZ-104 – LAB 06

Implement Network Traffic Management (Conceptual Completion)

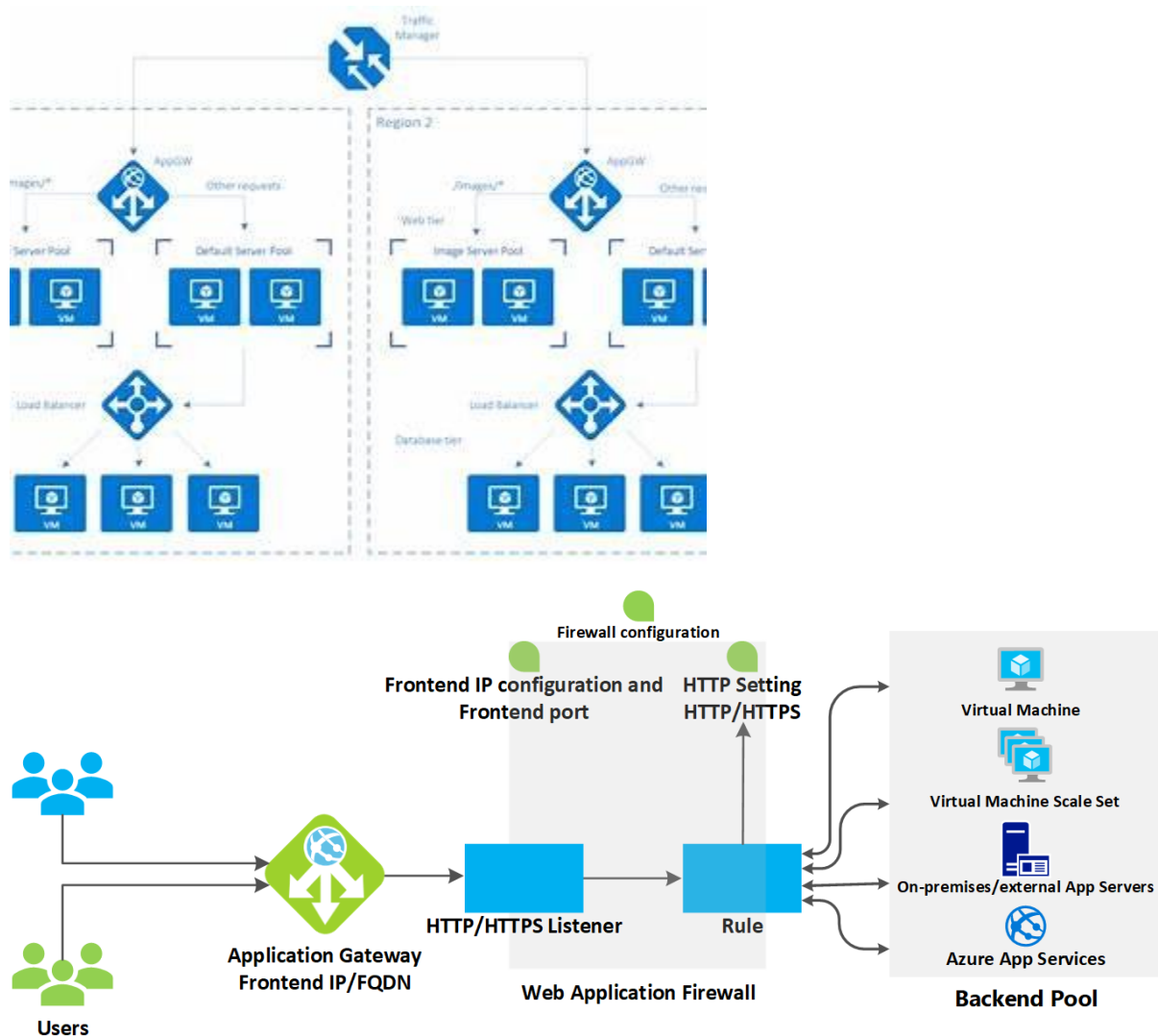
What Lab 06 is REALLY testing

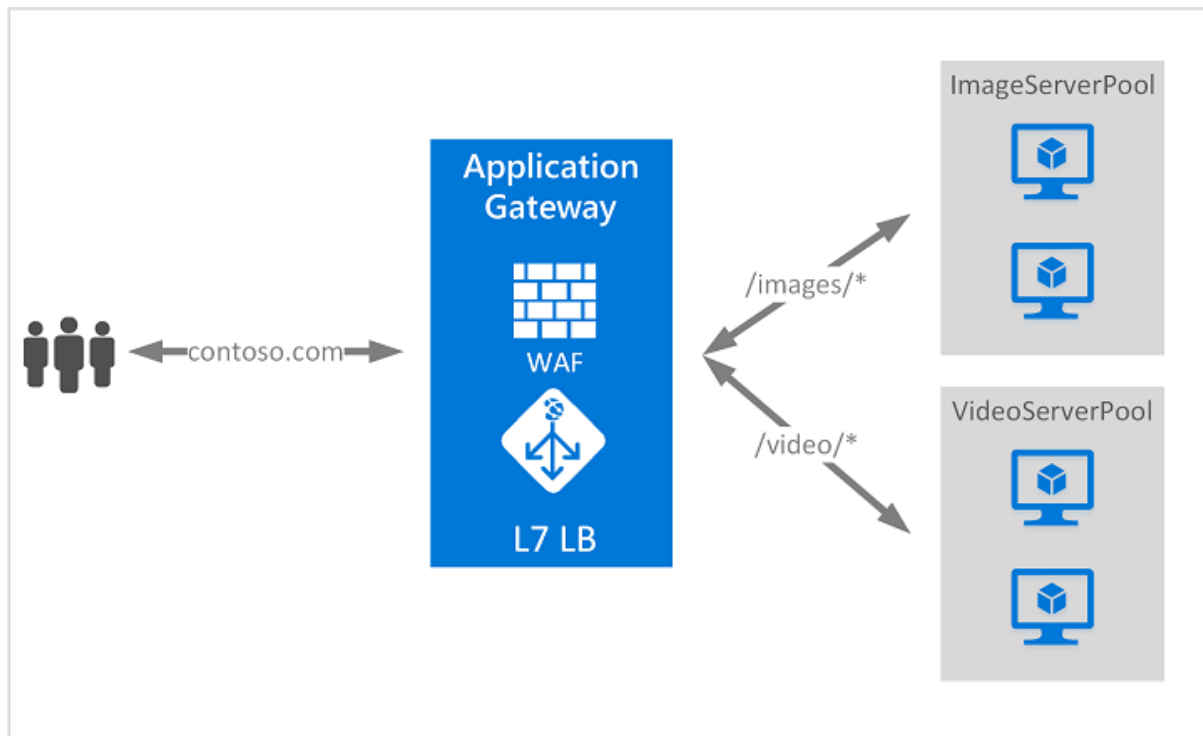
Lab 06 is **not** about VM deployment.

It tests whether you understand:

- 1 How traffic enters Azure
- 2 How Azure distributes traffic
- 3 When to use which load-balancing service

BIG PICTURE FLOW





User → Azure → Load Balancer / Application Gateway → VMs

◆ PART 1: AZURE LOAD BALANCER (LAYER 4)

◆ What Azure Load Balancer is

A **Layer-4 (TCP/UDP)** load balancer that distributes traffic based on **IP + Port**.

It DOES:

- ✓ Distribute traffic
- ✓ Provide high availability
- ✓ Perform health probes

It DOES NOT:

- ✗ Inspect URLs
- ✗ Understand HTTP paths
- ✗ Perform SSL termination

◆ Key Components (Understand this clearly)

Component	Purpose
Frontend IP	Public or private entry point
Backend pool	VMs that receive traffic
Health probe	Checks VM health
Load-balancing rule	Defines traffic flow

◆ How traffic flows (Load Balancer)

Client → Frontend IP

→ Rule (Port 80)

→ Backend pool

→ Healthy VM

Azure rotates traffic across VMs.

◆ Why refresh shows different VMs

Because:

- Session persistence = None
- Load balancer distributes each request
- No stickiness

This proves **true L4 balancing**.

◆ PART 2: AZURE APPLICATION GATEWAY (LAYER 7)

◆ What Application Gateway is

A **Layer-7 (HTTP/HTTPS)** load balancer with **intelligent routing**.

It DOES:

- ✓ Read URLs
- ✓ Perform path-based routing

- ✓ Terminate SSL
 - ✓ Support WAF
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◆ Why Application Gateway needs a dedicated subnet

Because:

- It's a managed service
- Azure controls scaling & routing
- Minimum subnet size: **/27**

This is **non-negotiable**.

◆ Path-Based Routing (MOST IMPORTANT)

Application Gateway can route like this:

/image/* → Image servers

/video/* → Video servers

This is **IMPOSSIBLE** with Azure Load Balancer.

◆ How traffic flows (Application Gateway)

Client → App Gateway

→ Listener (Port 80/443)

→ Rule

→ Backend pool

◆ LOAD BALANCER vs APPLICATION GATEWAY (MEMORIZE)

Feature	Load Balancer	Application Gateway
OSI Layer	4	7
URL-aware	✗	✓
Path routing	✗	✓

Feature	Load Balancer Application Gateway	
SSL termination	✗	✓
WAF	✗	✓
Cost	Low	Higher

WHEN TO USE WHAT (INTERVIEW GOLD)

Use Azure Load Balancer when:

- You need raw performance
- You don't care about HTTP content
- TCP/UDP workloads

Use Application Gateway when:

- You need intelligent routing
 - You host web apps
 - You need SSL/WAF
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WHY LAB 06 USES BOTH

Microsoft wants you to see:

- Layer-4 vs Layer-7 difference
 - Why **both** exist
 - Why they are **not interchangeable**
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COMMON INTERVIEW QUESTIONS & ANSWERS

Q1: Can Azure Load Balancer do path-based routing?

A: No. It operates at Layer 4 and cannot inspect URLs.

Q2: Why does Application Gateway cost more?

A: Because it performs deep packet inspection, SSL termination, and advanced routing.

Q3: Can Application Gateway replace Load Balancer?

A: No. They serve different purposes and layers.

Q4: Why is a health probe required?

A: To ensure traffic is only sent to healthy backends.

Q5: Why can't App Gateway share a subnet?

A: Azure requires a dedicated subnet to manage scaling and routing.
